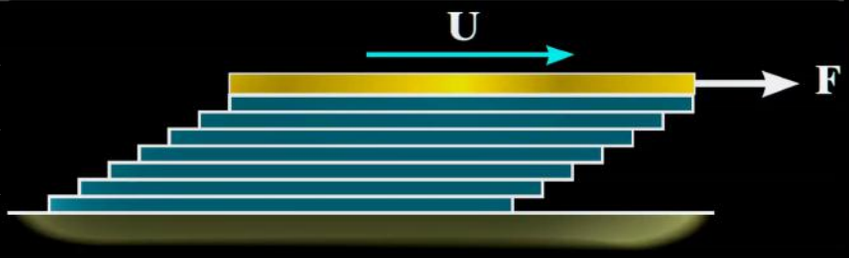
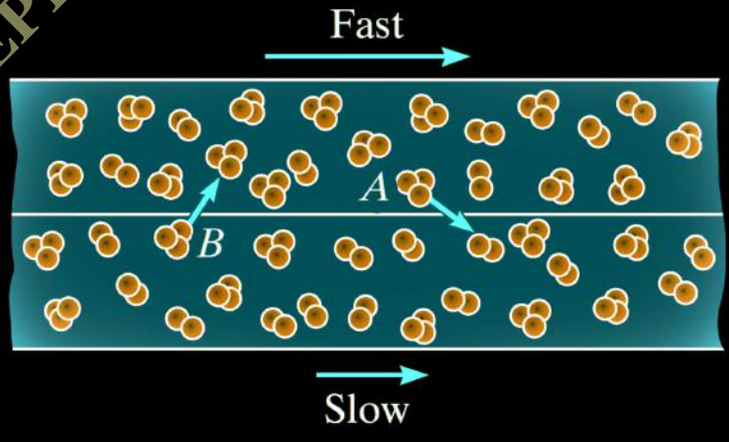
Viscosity:



Whenever a fluid is subjected to a shearing force, the fluid begins to flow. During the process of flowing, the layers of fluid slide over each other and a velocity gradient is produced. (By velocity gradient we mean rate of change of velocity in a direction perpendicular to the shearing force). The velocity gradient depends on the "stickiness" of the fluid, on how strongly the motion of one layer affects the motion of next layer. Greater the stickiness between the layers, smaller is the velocity gradient because layers want to "stick together". Thus a stickier liquid will not flow that freely and offer a greater resistance to flow. The quality of stickiness or resistance to flow is called viscosity.

Physical Cause Of Viscosity:



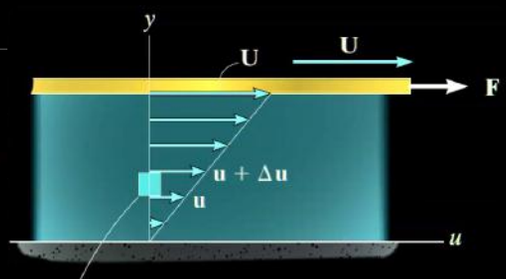
The resistance to flow that gives rise to viscosity can be understood by considering two layers of fluid sliding relative to each other. Since the molecules composing the fluid are always in continuous motion, then if molecule A in the faster top layer happens to travel down to the slower bottom layer, it will have a component of relative motion to the right. Collisions that occur with any slower moving molecule in bottom layer will cause it to be pushed along due to momentum exchange with A. Similarly if molecule B in the Bottom layer moves upwards, it will retard a faster moving molecule through momentum exchange thus trying to slow down the upper layer. Thus the upper layer is a bit "reluctant" to flow freely upon being sheared and this "reluctance" is a property of liquid, governed by particle sizes and nature of intermolecular interactions.

YOUTUBE.COM/@THEPHYSICSSHELTER

Shear Stress:



Distortion of fluid elements due to shear
(a)



Velocity distribution within a thin fluid layer
(b)



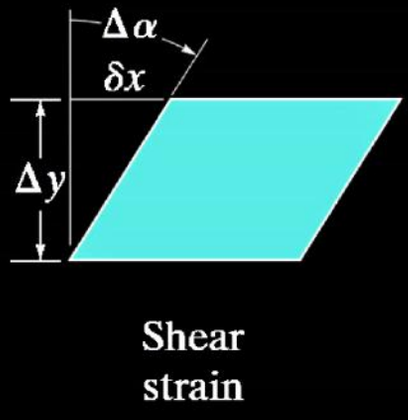
Shear stress causes Shear strain
(c)

Consider a fluid confined between a fixed surface and a very wide horizontal plate. When a force F is applied to the plate, it causes fluid elements to deform. After a brief period of acceleration, the viscous forces bring the plate in to equilibrium such that the plate begins to move with a constant velocity U. During this motion, the adhesive forces between fixed surface and liquid molecules keep the lowest molecules at rest wrt fixed surface i.e "no slip" condition. Similarly adhesive forces between lower surface of top plate and liquid molecules in contact with it move along with plate without slipping. Between these two surfaces, very thin layers of fluid are dragged along creating a velocity profile as shown in figure b. The layers will move parallel to the dragging plate. In steady state, shear force acting on top and bottom surface of any layer must be same (∵ no acceleration). If ΔF is the force that acts on top and bottom area ΔA of any element then, shearing stress is defined as force per unit area i.e :

$$\tau = \lim_{\Delta A \to 0} \frac{\Delta F}{\Delta A}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Shear Strain:



Shear strain

Since the fluid flows, the shear stresses will cause each element to deform in to a parallelogram and during a small time interval, δt, the resulting deformation is defined by shear strain specified by a small angle $\delta\alpha$ where $\delta\alpha \approx \tan\delta\alpha = \delta x/\delta y$

Unlike the case of solids, under action of shearing forces, the fluids continue to flow so instead of shear strain, time rate of change of shear strain is more important. If the top of the element has a velocity δu higher than bottom layer then $\delta x = \delta u\,\delta t$, thus, $\delta\alpha = \delta u\,\delta t/\delta y$

$$\Rightarrow \frac{\delta\alpha}{\delta t} = \frac{\delta u}{\delta y} \leftarrow \text{Strain rate.}$$

Now in the limit as δt and δy go to zero, we can write:

$$\frac{d\alpha}{dt} = \frac{du}{dy}$$, Here the RHS term $\frac{du}{dy}$ is called velocity gradient.

Newton's Law Of Viscosity:

In late 17th century, Newton proposed that shear stress in a fluid is directly proportional to the strain rate on the velocity gradient i.e.

$$\boxed{\tau = \mu \frac{du}{dy}} \leftarrow \text{Newton's law of viscosity.}$$

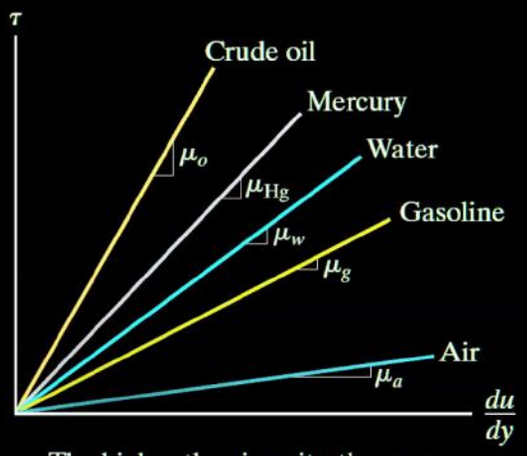
Here μ is called coefficient of viscosity (or dynamic or absolute viscosity). We will simply call it viscosity. Its SI unit is Ns/m^2 or Pascal Seconds. Its CGS unit is poise and SI unit is alternatively called deca poise i.e 10 poise.

10 Poise = 1 deca poise. Thus whenever viscosity is specified in Poise, we must always multiply it by 0.1 to convert to SI units.

YOUTUBE.COM/@THEPHYSICSSHELTER

NEWTONIAN FLUIDS:

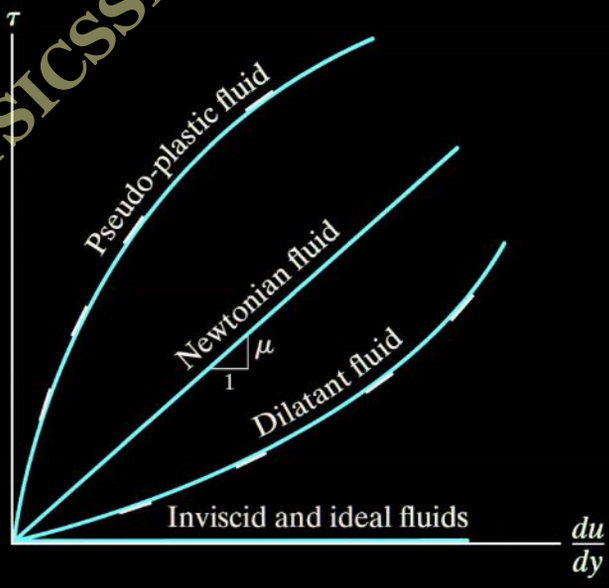
Any fluid that has a constant viscosity irrespective of value of shear stress or strain rate is called a Newtonian fluid. For such fluids graph between shear stress and velocity gradient is a straight line. Notice how the slope (viscosity) in the figure increases from air (which has very low viscosity) to crude oil (highly viscous). Higher the viscosity, more is the resistance to fluid flow.



The higher the viscosity, the more difficult it is for a fluid to flow

NON-NEWTONIAN FLUIDS:

Fluids which exhibit a non linear behavior between shear stress and velocity gradient are called non-newtonian fluids. If the viscosity (slope of graph) increases with increase in velocity gradient, the fluids are called dilatant (e.g. thick sugar syrup (Chaashnee). If the viscosity of the fluid decreases with increase in velocity gradient, the fluid is called pseudo-plastic fluid (e.g. blood, ketchup, gelatin and milk). Quicksand is another example of pseudo plastic fluid, that's why moving one's limbs faster makes one sink faster due to decreased viscosity. There can also be other substances which show a behavior intermediate between solids and liquids such as toothpaste or wet cement (these are studied in Rheology and not fluid mechanics).

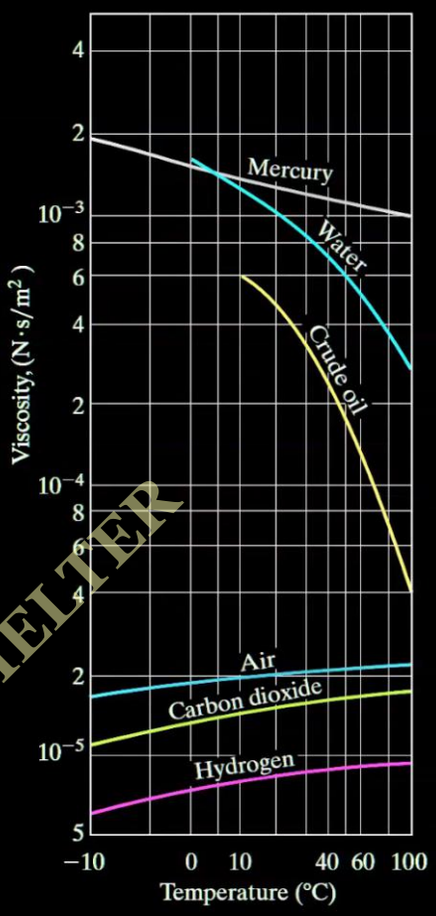


YOUTUBE.COM/@THEPHYSICSSHELTER

Effect Of Temperature On Viscosity:

With increase in temperature, the viscosity of liquids decreases because increase in temperature causes more thermal vibration and hence breaking the molecular bonds, allowing the molecules to more easily slip over each other. For liquids the viscosity curves are modeled using Andrade's equation i.e $\mu = Be^{C/T}$.

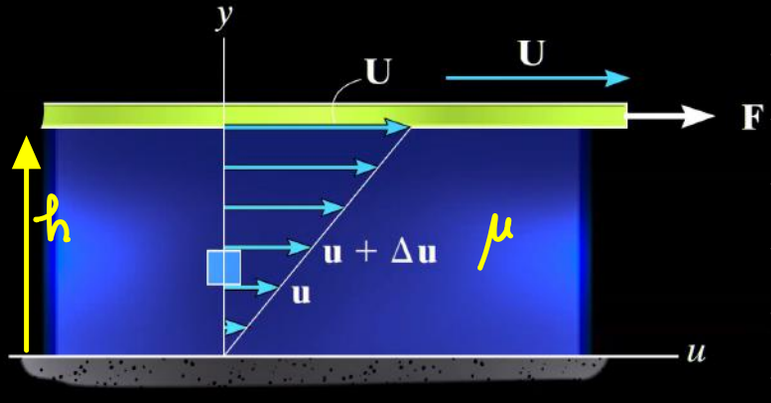
In case of gases, the viscosity increases with temperature. This is because increase in temperature increases the molecular motion in gases which increases momentum exchange between successive layers (molecular attraction forces are very small in gases unlike in liquids). The viscosity of gases is modeled using Sutherland's equation i.e $\mu = \frac{BT^{3/2}}{T+C}$



Temperature increase causes an increase in the viscosity of a gas, but a decrease in the viscosity of a liquid

Kinematic Viscosity: The ratio of dynamic viscosity (or simply viscosity) to the density of a fluid is called kinematic viscosity i.e $\boxed{\nu = \mu/\rho}$

Example: A wide plate of area A is being pulled by a force F on a thin oil film of thickness h and viscosity μ. Find the steady state velocity of the plate.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Since fluid layers are not accelerated, each layer must be experiencing same shear stress from top and bottom (though shearing forces are equal and opposite). Thus shear stress on every layer must be same as that on top layer, i.e $\tau = F/A$. Now using Newton's law of viscosity: $\tau = \mu \, du/dy$ thus:

$$\frac{F}{A} = \mu \frac{du}{dy} \Rightarrow \frac{F}{A} dy = \mu \, du \Rightarrow \frac{F}{A} \int_0^h dy = \mu \int_0^U du$$

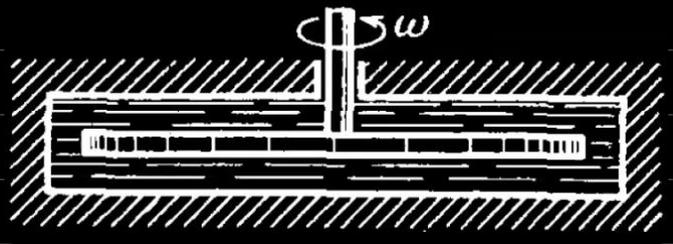
$$\Rightarrow U = Fh/A\mu \quad \text{Ans.}$$

→ Thus in this type of situations $\frac{du}{dy} = \frac{U}{h}$ and we can use this directly (velocity gradient is constant).

→ A similar velocity gradient may be assumed whenever a thin layer of liquid is trapped between a stationary and moving flat surfaces.

EXAMPLE: IRODOV 1.331:

A thin horizontal disc of radius R is located in a cylindrical cavity filled with oil whose viscosity is η. The clearance between the disc and horizontal planes of the cavity is h. Find the power developed by viscous forces on the disc when it rotates with angular velocity ω.

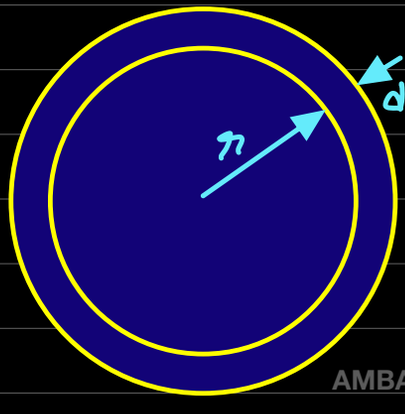


Solution: Velocity gradient between disc surface and cavity planes at a distance r is $\omega r/h$. Thus shear stress is

$$\tau = \eta \frac{du}{dy} = \eta \frac{\omega r}{h}.$$

Now consider an annulus or disc between radii r and $r+dr$. The torque experienced by this is $dM = \tau \times 2\pi r dr \times r \times 2$

$$\Rightarrow dM = \frac{4\pi\eta}{h} r^3 dr \, \omega$$ (Note factor of 2 due to both upper and lower surfaces).

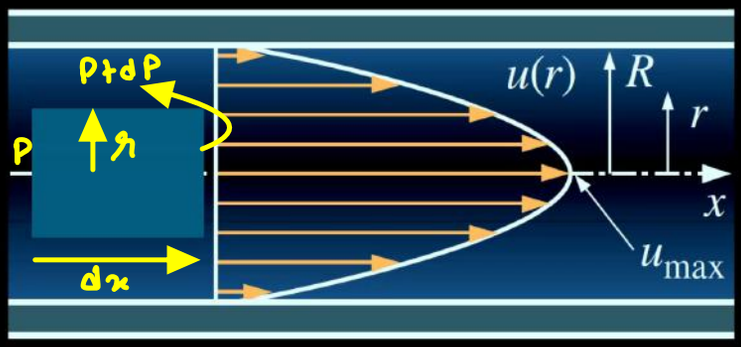


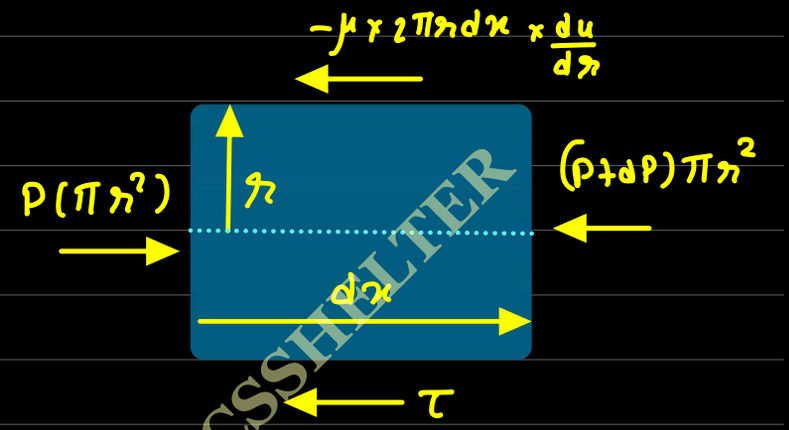
YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Thus Net viscous torque, $M = \frac{4\pi\eta\omega}{h}\int_0^R r^3 dr = \frac{\pi\eta\omega R^4}{h}$

Now power developed by viscous torque, $P = M\omega\cos 180^\circ = -\frac{\pi R^4\eta\omega^2}{h}$ Ans.

VELOCITY PROFILE IN A HORIZONTAL CIRCULAR PIPE:





Consider FBD of a symmetric cylindrical flowing element. Since the flow is steady, net resultant of pressure and shearing forces on the fluid element must be zero. Thus:

$$P\pi r^2 = (P+dP)\pi r^2 - \mu \times 2\pi r dx (du/dr)$$

$$\Rightarrow \frac{1}{2\mu}\left(\frac{dP}{dx}\right) r dr = du \text{, now @ } r = R,\ u = 0\text{, thus integrating:}$$

$$\frac{1}{2\mu}\frac{dP}{dx}\int_R^r r dr = \int_0^u du \Rightarrow \boxed{u = \frac{-1}{4\mu}\frac{dP}{dx}(R^2 - r^2)}$$

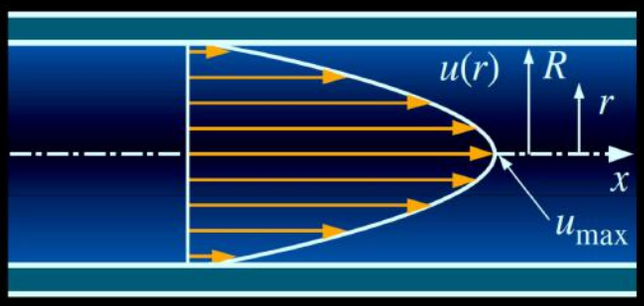
If we define $\Delta P = P_{inlet} - P_{outlet}$, $\frac{dP}{dx} = \frac{-\Delta P}{l}$ where l is pipe length.

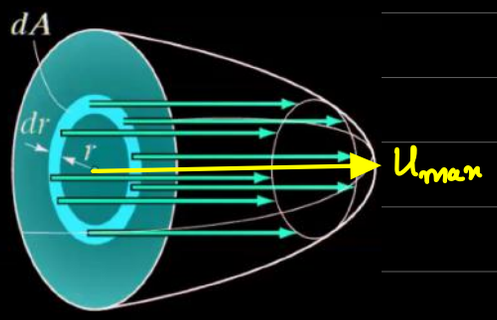
Thus $\boxed{u = \frac{\Delta P}{4\mu l}(R^2 - r^2)}$

→ Max speed occurs at $r = 0$ i.e $u_{max} = \frac{\Delta P R^2}{4\mu l}$

YOUTUBE.COM/@THEPHYSICSSHELTER

Volumetric Flow Rate (Hagen Poiseuille Equation):





We can either find the volumetric flow rate by integrating the elemental flows over annular sections or alternatively we can find the volume of velocity parabaloid as discussed in chapter on fluid dynamics (Bernoulli). Thus volumetric flow rate:

$$Q = \frac{1}{2} \times \pi R^2 \times U_{max} = \frac{\Delta P}{8\mu l} \pi R^4$$

Thus $\boxed{Q = \frac{\Delta P}{8\mu l} \pi R^4}$ ← Hagen Poiseuille Equation

Flow Resistance:

Just as in case of electric current, the current is the ratio of voltage drop (ΔV) to resistance similarly here we visualize the "volume current" or Volumetric flow rate as ratio of pressure drop to the flow resistance. Rearranging Hagen Poiseuille equation:

$Q = \Delta P / [8\mu l / \pi R^4]$, thus the term $8\mu l / \pi R^4$ acts like flow resistance. Thus $\boxed{R_{flow} = \frac{8\mu l}{\pi R^4}}$

→ The idea of flow resistance can be used for solving fluid flow circuits just like electrical circuits (Ideas like series and parallel combinations can be extended to fluid circuits).

YOUTUBE.COM/@THEPHYSICSSHELTER

AVERAGE FLOW VELOCITY: The ratio of volumetric flow rate to cross sectional area is called average flow velocity. Thus,

$$u_{avg} = \frac{\Delta P}{8\mu l} \cdot R^2 \quad \text{also} \quad u_{max} = \frac{\Delta P R^2}{4\mu l}$$

Thus $\boxed{u_{avg} = u_{max}/2}$

HEAD LOSS PER UNIT LENGTH: When fluid is flowing along a horizontal pipe, the velocity head and gravity head remain constant however pressure head is lost due to negative work done by viscous forces. The head loss per unit length is given as $\frac{1}{\rho g}\left(\frac{\Delta P}{l}\right)$.

Thus head loss per unit length of pipe is $\frac{h}{l} = \frac{8\mu Q}{\rho g \pi R^4}$.

This head loss may be accounted for in the adjusted Bernoulli's equation. While we have proved our result for horizontal pipes, same result also holds for laminar flows in inclined or vertical pipes as well.

EXAMPLE: PATHFINDER: BUILD: 32:



A pump circulates water at a time rate q_0 through a straight pipe of length l. Now a ring-shape section of mean radius r is connected in the middle of the pipe. Inner section of the ring is equal to that of the pipe as shown in the figure. If the same pump is connected to this new pipe, how much time rate q of water flow will the pump maintain? Assume that the pressure difference at the ends of the tube remains unchanged.

Solution: For same pressure drop, the flow rates will be in inverse ratio of resistances. Since cross sectional area of every pipe section is same, we can assume the flow resistance of any pipe to be proportional to its length with some proportionality constant k.

Thus initial resistance: $R_1 = kl$

Final resistance: $R_2 = k(l-2r)$ series $(k\pi r \,||\, k\pi r)$

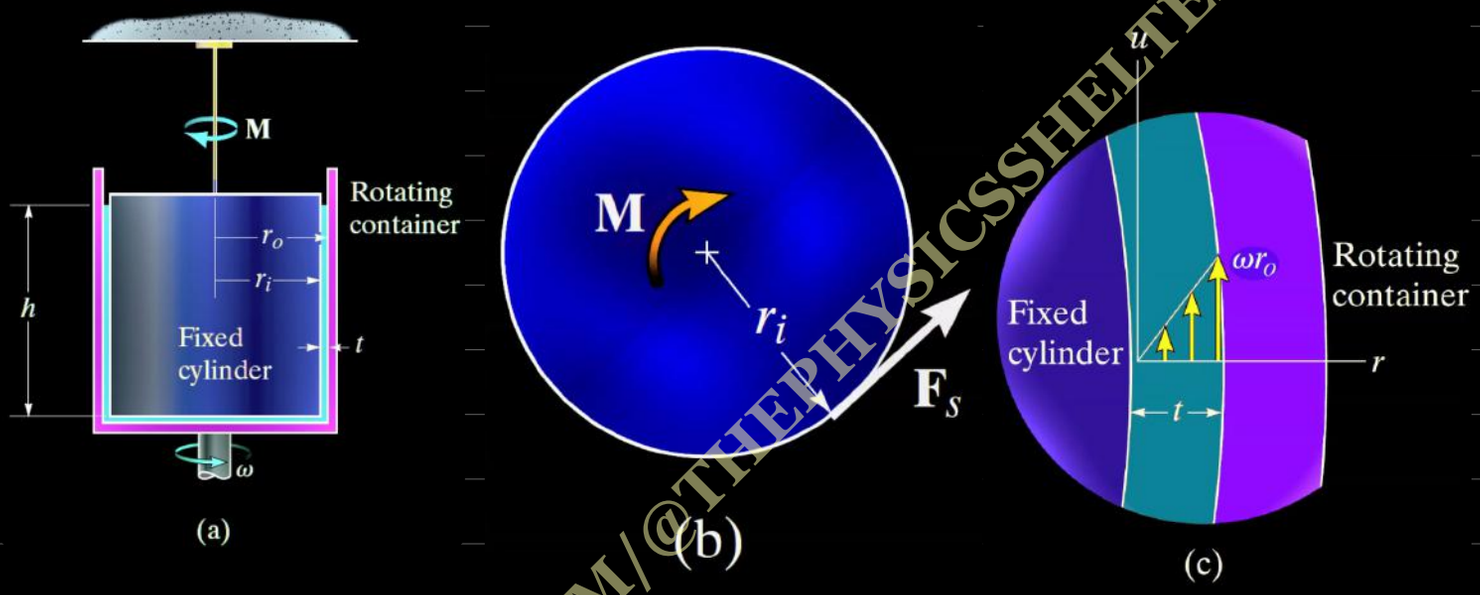
$$= k(l-2r) + \frac{k\pi r}{2}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Thus final flow rate, $q_2 = q_0 \frac{R_1}{R_2} = \frac{q_{vo}\, l}{[l - 2r + \pi r/2]}$ Ans.

→ In our simplistic analysis we have ignored head losses around sharp bends, however in practice, engineers do take in to account head losses at every sharp bend.

ROTATIONAL VISCOMETER WITH THIN FLUID LAYER: (BROOKFIELD VISCOMETER):



In this viscometer, the fluid is contained in a very thin gap between a fixed cylinder and a rotating cylinder. Because of shearing forces, inner cylinder experiences a shearing torque which is balanced by a torsional spring. Knowing the torque offered by the rotating liquid, one can estimate the viscosity of the fluid. Since the fluid layer is very thin, the flow can be modeled as the flow between a fixed flat surface and a moving plate. Referring to the figure, we can thus write the torque of shearing force on inner cylinder as:

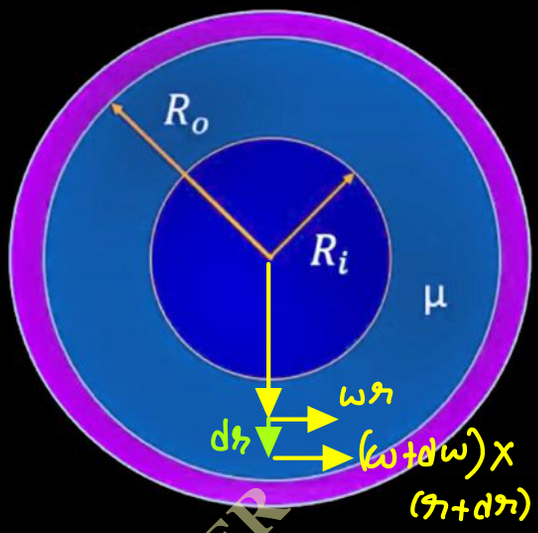
$$M = 2\pi r_i h \times \mu \times \left(\frac{\omega r_o}{t}\right) \times r_i$$

Thus $$\mu = \frac{M t}{2\pi \omega r_i^2 r_o h}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

COUETTE VISCOMETER:

This is similar to the Brookfield viscometer except that here fluid layer between inner cylinder and outer cylinder need not be thin.



VELOCITY GRADIENT IN COUETTE VISCOMETER:

Consider two layers of fluid located at r and r+dr rotating with angular velocity ω and ω+dω. From rotating frame of the inner layer, outer layer seems to rotate around it with an angular velocity dω. Thus velocity gradient (for purpose of calculating shear strain rate) is simply r dω/dr.

SHEAR STRESS IN COUETTE VISCOMETER:

Recall that $\tau = \mu \frac{du}{dy}$ thus in situation of couette viscometer,

$$\boxed{\tau = \mu r \frac{d\omega}{dr}}$$

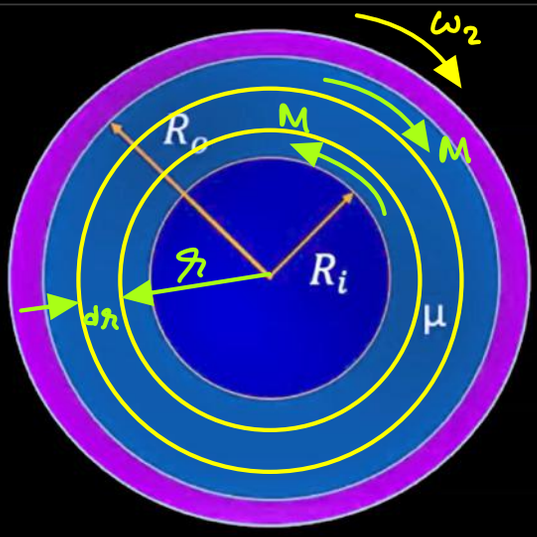
EXAMPLE: IRODOV 1.333:

A fluid with viscosity η fills the space between two long co-axial cylinders of radii R_1 and R_2, with $R_1 < R_2$. The inner cylinder is stationary while the outer one is rotated with a constant angular velocity ω_2. The fluid flow is laminar. Taking into account that the friction force acting on a unit area of a cylindrical surface of radius r is defined by the formula $\tau = \mu r(\partial\omega/\partial r)$, find :

(a) the angular velocity of the rotating fluid as a function of radius r;

(b) the moment of the friction forces acting on a unit length of the outer cylinder.

YOUTUBE.COM/@THEPHYSICSSHELTER



Solution: This is standard situation of a couette viscometer. Here since angular momentum of any layer remains constant, the torque of shearing forces on inner and outer surface of any layer must be equal and opposite and therefore magnitude of torque on every cylindrical surface due to shearing forces must be constant (say M).

Now $\tau = \mu r \frac{d\omega}{dr}$, $\Rightarrow$ considering a depth l:

$$M = \tau \times 2\pi r l \times r = 2\pi\mu l r^3 \frac{d\omega}{dr}$$

$$\Rightarrow \int_0^{\omega} d\omega = \frac{M}{2\pi\mu l} \int_{R_1}^{r} \frac{dr}{r^3} \quad \Rightarrow \quad \omega = \frac{M}{4\pi\mu l}\left[\frac{1}{R_1^2} - \frac{1}{r^2}\right] \quad (1)$$

Also at $r = R_2$, $\omega = \omega_2$, thus $\omega_2 = \frac{M}{4\pi\mu l}\left[\frac{1}{R_1^2} - \frac{1}{R_2^2}\right]$ (2)

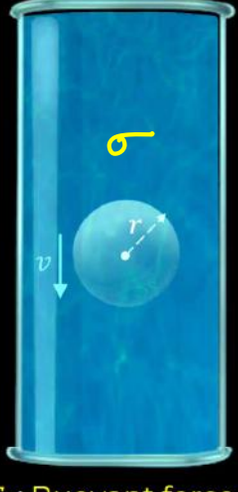
Solving (1) & (2):

$$\omega = \frac{R_2^2 R_1^2}{(R_2^2 - R_1^2)}\left(\frac{1}{R_1^2} - \frac{1}{r^2}\right)\omega_2 \text{ Ans,} \quad \frac{M}{l} = 4\pi\mu\omega_2 \frac{R_2^2 R_1^2}{R_2^2 - R_1^2} \text{ Ans.}$$

Viscous Drag On A Moving Sphere (Laminar Flow): Stoke's Law:

If a sphere of radius r is moving in a wide pool with a velocity v and the viscosity of the fluid flowing around the sphere is μ then the drag force experienced by the sphere is given as:

$$\boxed{F_v = 6\pi\mu r v}$$



F_B: Buoyant force

YOUTUBE.COM/@THEPHYSICSSHELTER

→ If the pool is not sufficiently wide, then drag force is still greater because fluid gets narrower passage around sphere leading to greater velocity gradients.

TERMINAL VELOCITY:

CASE I: Ball denser than surrounding liquid: As the ball sinks, its velocity increases thus increasing the drag force. When the drag force becomes sufficient to balance the resultant of Buoyant force and the weight, the ball will start moving with a constant velocity known as terminal velocity.

Now let us consider FBD of the ball:

$mg = F_B + F_v$ (F_B: Buoyant force, $F_v = 6\pi\mu r v$)



Thus $\frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi\mu r v$

σ: Density of liquid

$\Rightarrow$ $\boxed{v = \frac{2}{9\mu} r^2 g (\rho - \sigma)}$ downwards

CASE II: $\sigma > \rho$

Suppose a wooden ball is released from floor of an ocean or a lake then it rises above. As its upward speed increases, it experiences an increased downward drag. When the drag force is sufficiently large to balance the resultant of weight and Buoyant force, the ball moves upwards with a constant velocity known as terminal velocity. Here FBD will be similar to the previous case except that drag force will reverse its direction. Thus:

$$\rho \times \frac{4}{3}\pi r^3 g + 6\pi\mu r v = \sigma \times \frac{4}{3}\pi r^3 g$$

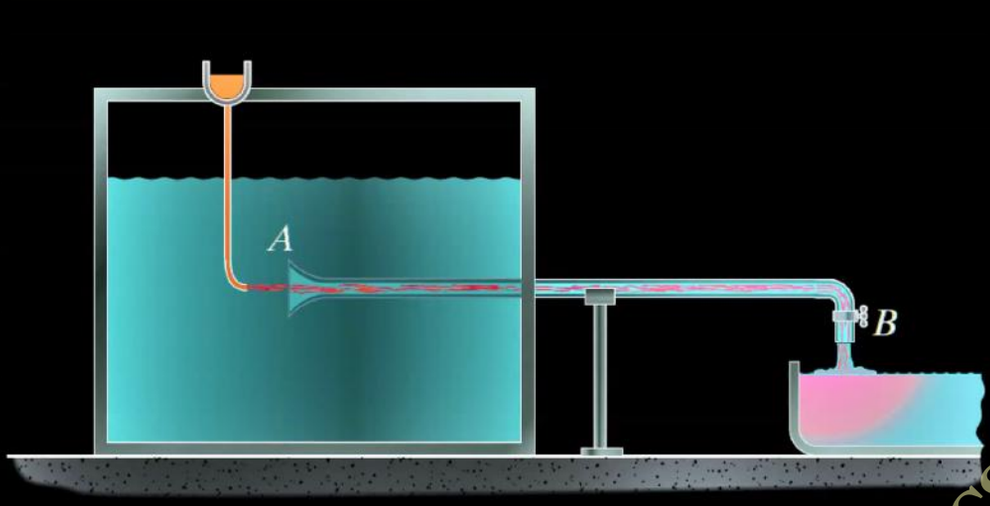
$\Rightarrow$ $\boxed{v = \frac{2}{9\mu} r^2 g (\sigma - \rho)}$ upwards.

YOUTUBE.COM/@THEPHYSICSSHELTER

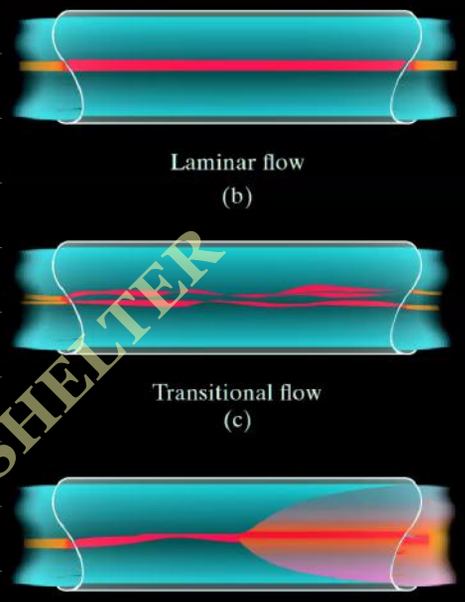
→ In both cases, magnitude of terminal velocity can be written as

$$v_t = \frac{2}{9\mu} r^2 g |\rho - \sigma|$$

REYNOLD'S NUMBER :



(a)



In 1883 Osborne Reynold used an apparatus similar to the one shown above in which a dye was systematically mixed in the flow stream at point A and flow rate was controlled by a knob at B. For slow flow rates the flow in the tube was observed to be laminar since the dye streak remained straight and uniform (figure b), by further opening the valve the streak began to break down as it underwent transitional flow. Finally a further increase in flow rate caused a turbulent flow because the dye became fully dispersed throughout water in the tube. He related these observations to flow velocity, density, viscosity and the diameter of the tube and came up with a criterion for classifying the flows as turbulent, transition or laminar. He proposed a dimensionless number known as Reynold's number which is defined as:

$$R_e = \frac{\rho v D}{\mu} = \frac{v D}{\nu}$$

Here v is the flow velocity

D: pipe diameter

ρ: fluid density, μ: viscosity

YOUTUBE.COM/@THEPHYSICSSHELTER

ν: kinematic viscosity (μ/ρ, Unit m^2/s, cas unit: stoke: cm^2/s).

According to NCERT if $R_e > 2000$ flow is turbulent, if $R_e < 1000$, flow is laminar and if $1000 < R_e < 2000$, flow is unsteady (transition)
(Though according to some engineering textbooks and Wikipedia:
$R_e < 2300 \Rightarrow$ laminar flow, $R_e > 2900 \Rightarrow$ turbulent flow and
$2300 < R_e < 2900 \Rightarrow$ transition flow).

→ Reynold's number is a way of comparing flows of various fluids undergoing similar flow geometry, for other types of geometries, other Reynold's numbers can be defined. For instance for flow around a sphere, Reynold's number could be defined using diameter of the sphere as characteristic dimension instead of the pipe diameter.

EXAMPLE: IRODOV 1.336:

The cross-sectional radius of a pipeline decreases gradually as $r = r_0 e^{-\alpha x}$, where $\alpha = 0.50\ \text{m}^{-1}$, x is the distance from the pipeline inlet. Find the ratio of Reynolds numbers for two cross-sections separated by $\Delta x = 3.2\ \text{m}$.

Solution: $R_e = \frac{\rho v D}{\mu}$ also by continuity, $v \times D^2 = \text{constant}$

$\Rightarrow v \propto \frac{1}{D^2} \Rightarrow R_e \propto \frac{1}{D} \propto \frac{1}{r}$, Thus $\frac{R_e + \Delta R_e}{R_e} = \frac{r_0 e^{-\alpha x}}{r_0 e^{-\alpha(x+\Delta x)}}$

$= e^{\alpha \Delta x} = e^{0.5 \times 3.2} = 4.95 \approx 5$. Ans.

EXAMPLE: IRODOV 1.338:

A lead sphere is steadily sinking in glycerin whose viscosity is equal to $\mu = 13.9$ P. What is the maximum diameter of the sphere at which the flow around that sphere still remains laminar? It is known that the transition to the turbulent flow corresponds to Reynolds number $\text{Re} = 0.5$. (Here the characteristic length is taken to be the sphere diameter.)

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: $Re = \frac{\rho v D}{\mu}$ also terminal velocity, $v = \frac{2}{9}\frac{g R^2 (\rho - \sigma)}{\mu}$

Thus, $Re = \frac{\rho D}{\mu} \cdot \frac{2}{9\mu} g R^2 [\rho - \sigma) = \frac{\rho D^3}{18\mu^2} g[\rho - \sigma]$

$\Rightarrow D = \left[\frac{18 Re\, \mu^2}{\rho \mu g [\rho - \sigma]}\right]^{1/3}$ Ans. ρ: Density of lead, σ: density of Glycerin

Putting the values from table, $D = 5$ mm. Ans.

EXAMPLE: IRODOV 1·337:

When a sphere of radius $r_1 = 1.2$ mm moves in glycerin, the laminar flow is observed if the velocity of the sphere does not exceed $v_1 = 23$ cm/s. At what minimum velocity v_2 of a sphere of radius $r_2 = 5.5$ cm will the flow in water become turbulent? The viscosities of glycerin and water are equal to $\mu_1 = 13.9$ P and $\mu_2 = 0.011$ P respectively.

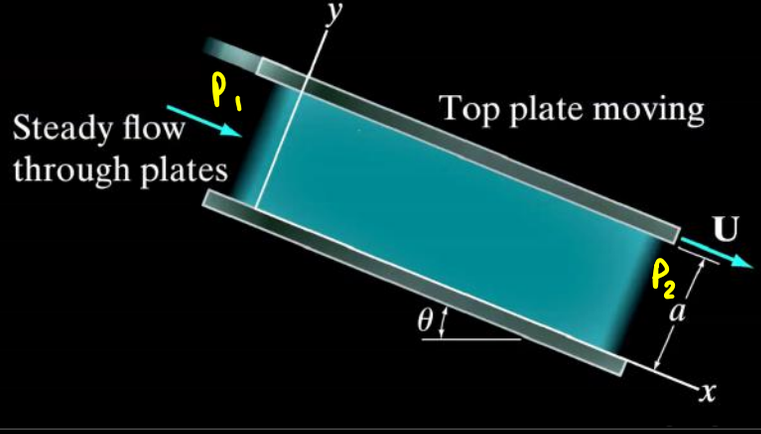
Solution: Equating critical reynold's numbers:

$$\frac{\rho_1 v_1 D_1}{\mu_1} = \frac{\rho_2 v_2 D_2}{\mu_2} \quad \Rightarrow \quad v_2 = \frac{\rho_1}{\rho_2} \cdot \frac{D_1}{D_2} \cdot \frac{\mu_2}{\mu_1} \times v_1$$

$\Rightarrow v_2 = \frac{\rho_1 r_1 \mu_2}{\rho_2 r_2 \mu_1} v_1$, putting the values, we get $v_2 = 5$ μm/sec Ans.

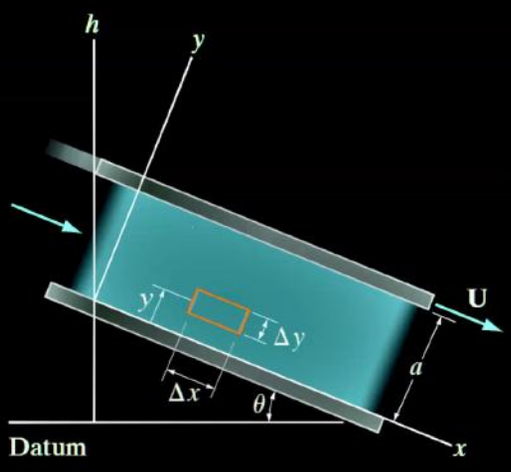
(In this question the author is not distinguishing between transition flow and turbulent flow).

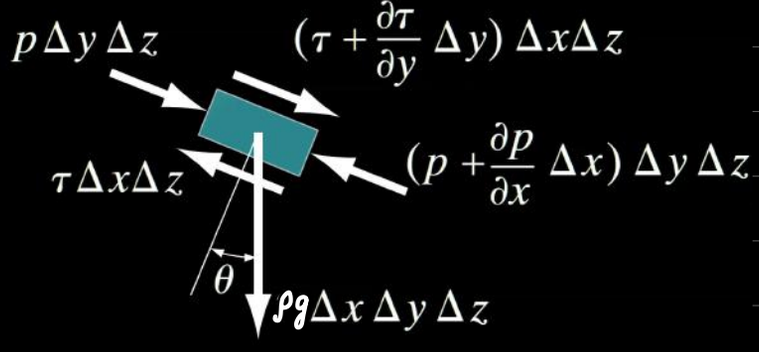
EXAMPLE: A fluid layer lies between a moving plate and a stationary plate. Develop a second order differential equation to solve for fluid velocity u as a function of y coordinate. Viscosity is μ and plates are very wide.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution:





Free-body diagram

Consider Free body diagram of a fluid element lying between y and y+dy.

$$\underbrace{p\,\Delta y\,\Delta z - \left(p + \frac{\partial p}{\partial x}\Delta x\right)\Delta y\,\Delta z}_{\text{Net pressure force}} + \underbrace{\left(\tau + \frac{\partial \tau}{\partial y}\Delta y\right)\Delta x\,\Delta z - \tau\,\Delta x\,\Delta z}_{\text{Net shear force}}$$

$$+\underbrace{\rho g \Delta x\,\Delta y\,\Delta z \sin\theta}_{\text{Weight}} = 0 + 0$$

Dividing throughout by $\Delta x \Delta y \Delta z$, we get:

$$-\frac{\partial p}{\partial x} + \frac{\partial \tau}{\partial y} + \rho g \sin\theta = 0$$

Also $\tau = \mu \frac{du}{dy} \Rightarrow \frac{\partial \tau}{\partial y} = \mu \frac{d^2u}{dy^2}$, Thus:

$$\mu \frac{d^2u}{dy^2} = -\rho g \sin\theta + \frac{\partial p}{\partial x}$$

For a spontaneously sliding plate, (No pressurised flow of fluid), $\frac{\partial p}{\partial x}$ can be taken as zero and we can solve for u considering velocity at upper and lower boundaries.

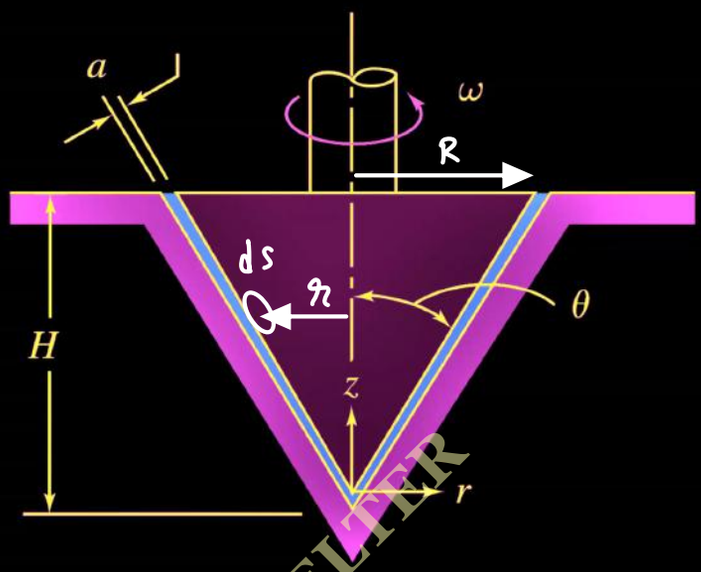
YOUTUBE.COM/@THEPHYSICSSHELTER

Viscous Torque On Spinning Objects (Constant Thin Film Thickness):

Consider a patch of area ds on surface of a rotating object, as shown in the figure. The viscous torque on the element is given as:

$$dM = \mu\left(\frac{\omega r}{a}\right) ds\, r = \frac{\mu\omega}{a} ds\, r^2$$

$$\Rightarrow M = \frac{\mu\omega}{a} \int ds\, r^2 \text{ , Now}$$



recognize that term inside the integral is area moment of inertia. Thus $\boxed{M = \frac{\mu\omega}{a} I}$ where I is Area moment of inertia about the spinning axis. For instance in the shown system, $I = A R^2/2 = \pi R^3 \sqrt{R^2+H^2}/2$

Thus, viscous torque, $M = \frac{\pi \mu\omega}{a} R^3 \sqrt{R^2+H^2}$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER